

ENM 5310 — Quiz 1: Solutions

Lectures 1–3 · 6 questions, 1 point each

Your quiz sheet was Version A or Version B — the questions are identical and in the same order, but the answer options are permuted. Find your version's column below.

Q	Topic	Version A	Version B
1	Monte Carlo error rate	C	D
2	Irreducible error in regression	B	C
3	Correlation vs. independence	C	B
4	Bayes' rule and base rates	B	C
5	Change of variables	A	D
6	KL divergence	D	A

1. Monte Carlo error rate. *Correct: 100× more samples; unchanged by dimension.*

The sample-average estimator $\hat{f}_N = \frac{1}{N} \sum_n f(x^{(n)})$ is unbiased with variance σ_f^2/N , so its RMSE is $\sigma_f/\sqrt{N}$. Cutting the error by a factor of 10 therefore requires $10^2 = 100$ times as many samples. Crucially, the exponent $-1/2$ contains no d : dimension affects the constant σ_f , never the rate. Compare a tensor-product quadrature rule of order r , whose error scales as $N^{-r/d}$ and so degrades exponentially with dimension. This asymmetry is the reason we will use sampling for high-dimensional integrals all semester, despite the slow rate. (*Lecture 1, §7.*)

If you chose the option that keeps the 100× but adds an exponential-in-dimension penalty: that is the quadrature intuition from numerical methods, and it does not apply here.

2. Irreducible error. *Correct: 0.25.*

For any g ,

$$\mathbb{E}[(y - g(x))^2] = \underbrace{\mathbb{E}[\mathbb{V}[y | x]]}_{\text{irreducible}} + \underbrace{\mathbb{E}[(\mathbb{E}[y | x] - g(x))^2]}_{\geq 0}.$$

The second term vanishes exactly when $g = \mathbb{E}[y | x] = f$, leaving $\mathbb{E}[\mathbb{V}[y|x]] = \text{Var}(\epsilon) = 0.25$. Note 0.25 is the noise **variance**, not the standard deviation. Two things follow, and they matter for every experiment you run in this course: no amount of data, capacity, or training reduces this floor, and you cannot tell a converged model from an underfit one unless you know what the floor is. (*Lecture 2, §5.*)

3. Correlation vs. independence. *Correct: $\rho_{XY} = 0$, but X and Y are dependent.*

$$\text{Cov}[X, X^2] = \mathbb{E}[X^3] - \mathbb{E}[X] \mathbb{E}[X^2] = 0,$$

since both odd moments vanish by symmetry. Yet Y is a deterministic function of X — maximal dependence with zero correlation. Correlation measures only the *linear* part of dependence, so symmetric, periodic, or threshold relationships are invisible to it. The implication runs one way: independence implies zero correlation, but zero correlation implies independence only when (X, Y) are **jointly Gaussian**. This is also why all four Anscombe datasets share $\rho = 0.816$ despite completely different structure. (*Lecture 2, §3.*)

4. Bayes' rule and base rates. *Correct: about 8%.*

$$p(\text{def} | +) = \frac{p(+ | \text{def}) p(\text{def})}{p(+ | \text{def}) p(\text{def}) + p(+ | \text{sound}) p(\text{sound})} = \frac{(0.9)(0.01)}{(0.9)(0.01) + (0.1)(0.99)} = \frac{0.009}{0.108} \approx 0.083.$$

In natural frequencies: of 10,000 parts, 100 are defective and 90 of those test positive; 9,900 are sound and 990 of *those* also test positive. So 1,080 positives, of which only 90 are real. When the base rate is low, false positives from the large negative population swamp true positives from the small positive one — which is why sensitivity and specificity alone never tell you how to act on a positive result, and why accuracy is a useless metric on imbalanced data. (*Lecture 2, §6.1.*)

5. Change of variables. *Correct: $p_Y(y) = 1/y$ on $[1, e]$.*

With $y = e^x$ the inverse is $g(y) = \ln y$, so

$$p_Y(y) = p_X(g(y)) \left| \frac{dg}{dy} \right| = 1 \cdot \frac{1}{y}, \quad y \in [1, e].$$

The Jacobian factor is not optional: a density is probability *per unit length*, so it must absorb the local stretching of the transformation. Always check normalization — here $\int_1^e y^{-1} dy = 1$. The option $p_Y = 1$ on $[1, e]$ (the Jacobian dropped) integrates to $e - 1 \approx 1.72$ and can be eliminated in seconds by that check alone. (*Lecture 3, §7.*)

6. KL divergence. *Correct: non-negative, zero only when $p = q$, measuring excess average surprise.*

$\text{KL}(q\|p) = \mathbb{H}(q, p) - \mathbb{H}[q] \geq 0$, by Jensen's inequality applied to the convex function $-\log$, with equality exactly when $p = q$. Read it as the extra nats per observation you incur by modeling data from q using p , over and above the irreducible uncertainty $\mathbb{H}[q]$ that no model can remove.

Two traps. The log-ratio $\log(q(x)/p(x))$ is negative over parts of the support — what is guaranteed non-negative is its **expectation**. And KL is asymmetric and violates the triangle inequality, so it is a divergence, not a distance; $\text{KL}(q\|p)$ and $\text{KL}(p\|q)$ answer different questions, a distinction that will matter when we reach variational inference. (*Lecture 3, §9.*)

To review: the relevant sections are cited after each solution, and Practice Sets 0.1–0.3 in the lecture notes cover the same material in more depth. Questions 2, 3, and 6 are load-bearing for the rest of the course — if any of those three is still unclear, come to office hours before we start on estimation.